Word Embeddings

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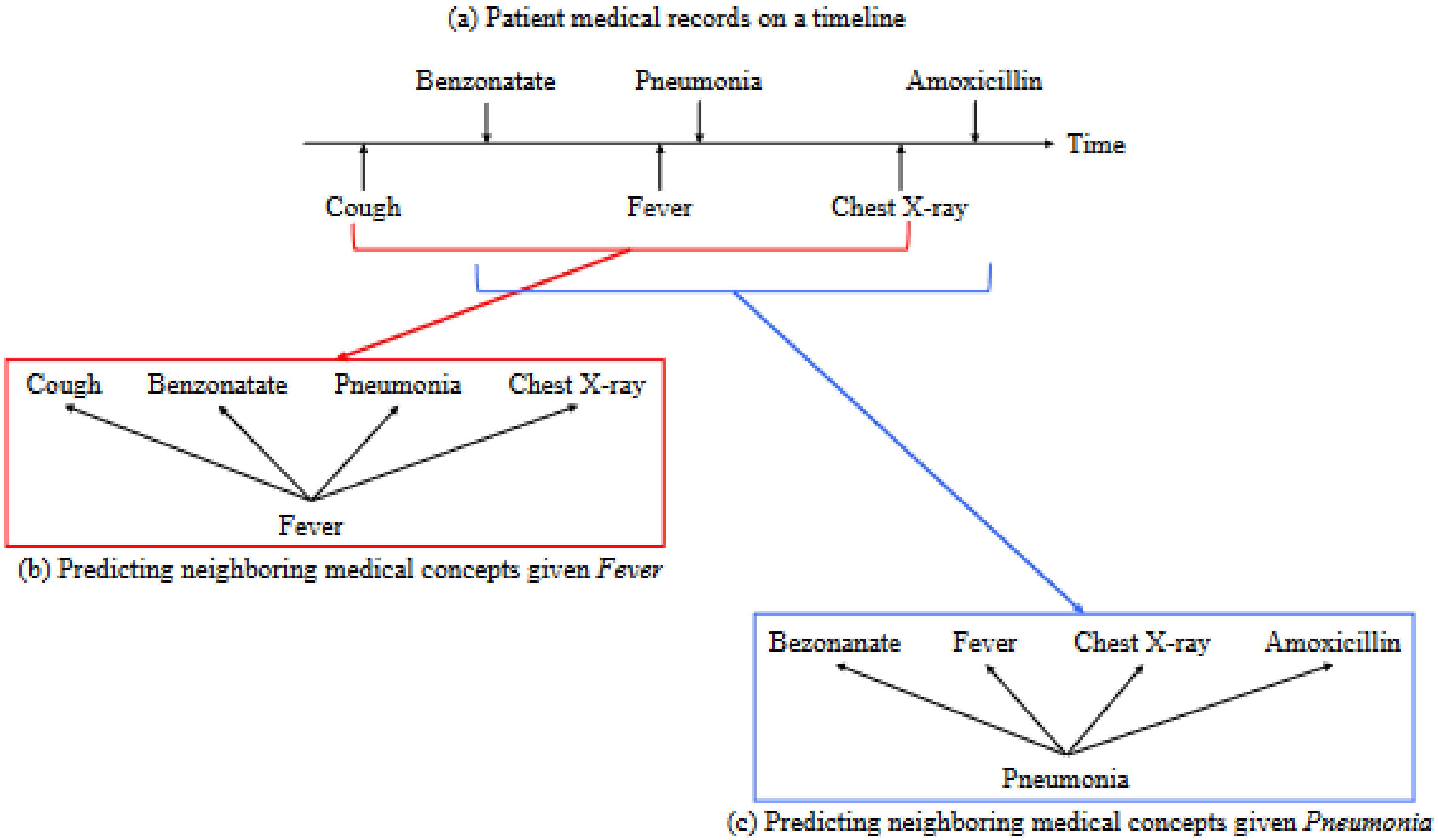

(a) Patient medical records on a timeline
Benzonatate
Pneumonia
Amoxicillin
Time
Cough
Fever
Chest X-ray
Cough
Benzonatate
Pneumonia
Chest X-ray
Fever
(b) Predicting neighboring medical concepts given Fever
Bezonanate
Fever
Chest X-ray
Amoxicillin
Pneumonia
(c) Predicting neighboring medical concepts given Pneumonia

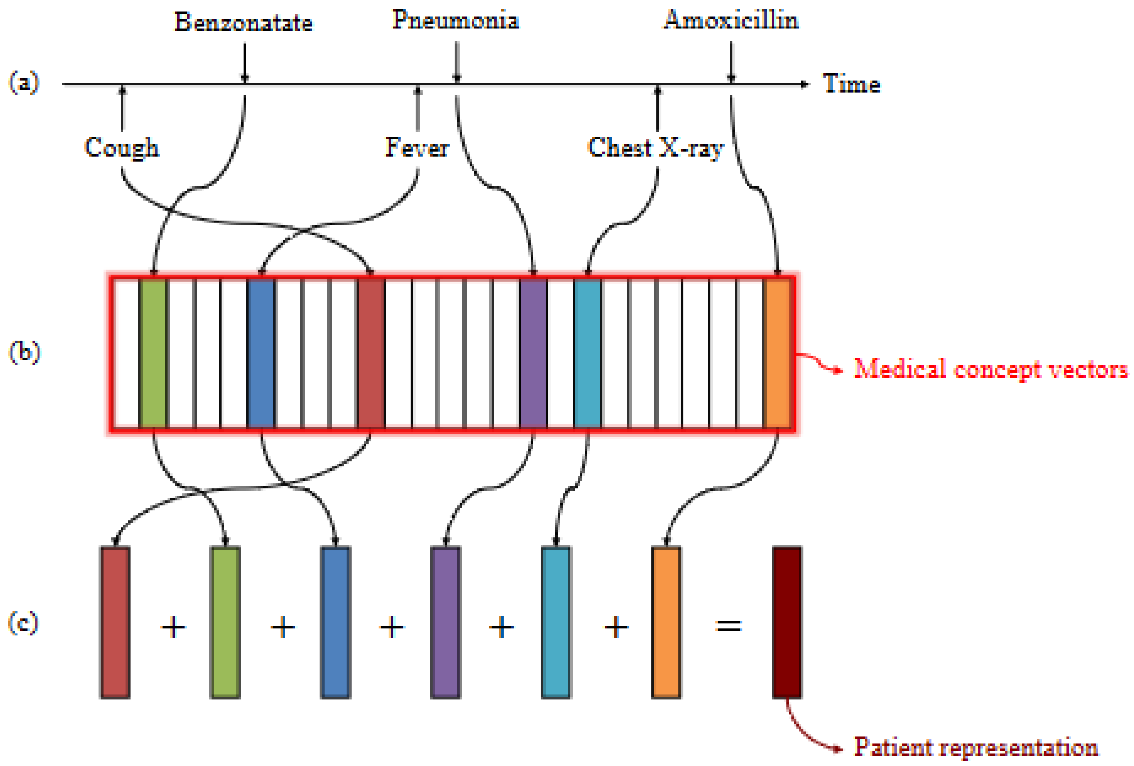
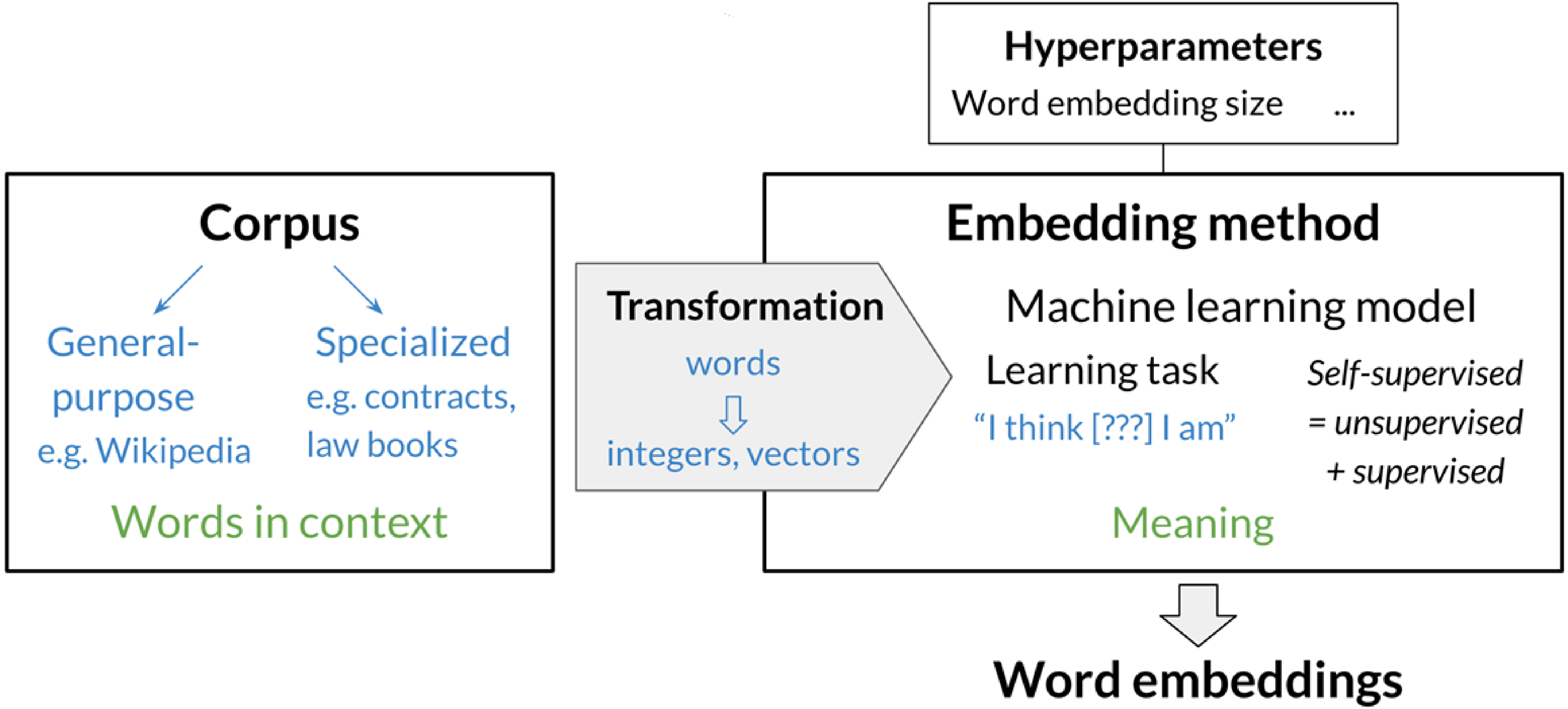
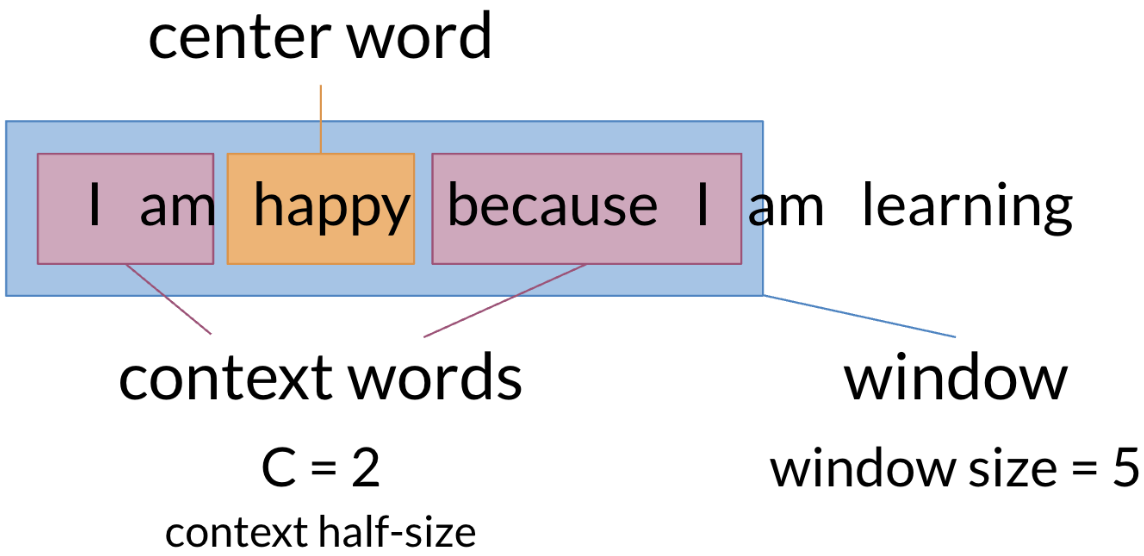
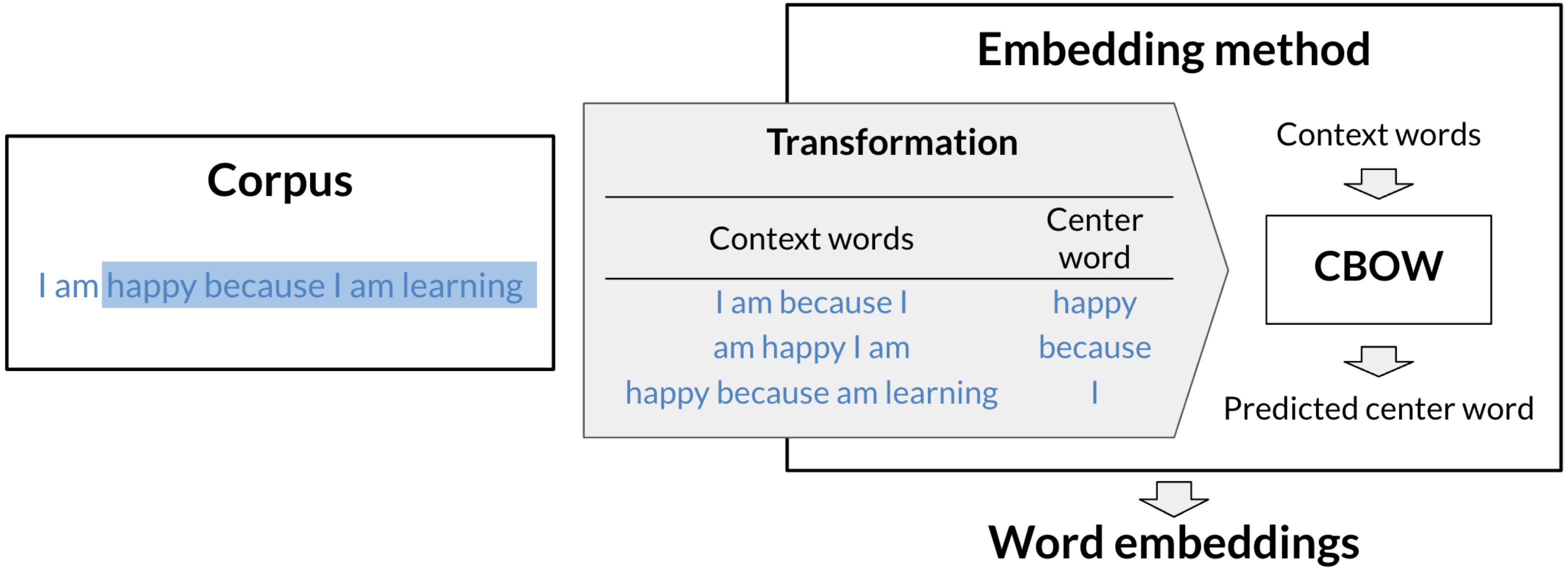


Figure 4. Patient representation construction. (a) represents a medical record of a patient on a timeline. (b) The medical concepts are represented as vectors using the trained medical concept vectors. (c) The patient is represented as a vector by summing all medical concept vectors.


Hyperparameters
Word embedding size ...
Corpus
General-purpose
e.g. Wikipedia
Specialized
e.g. contracts, law books
Words in context
Transformation
words
integers, vectors
Embedding method
Machine learning model
Learning task
"I think [???] I am"
Self-supervised
= unsupervised
+ supervised
Meaning
Word embeddings


center word
I am happy because I am learning
context words
window
C = 2
context half-size
window size = 5


Corpus
I am happy because I am learning
Transformation
Context words
Center word
I am because I
happy
am happy I am
because
happy because am learning
I
Embedding method
Context words
CBOW
Predicted center word
Word embeddings

HOW TO CREATE WORD EMBEDDINGS?

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CORPUS

- Dataset
- Text is the data type
- Preprocess (Vocabulary and Dictionary)

TRANSFORMATION

- Indices
- One-hot-encoding

- Letter case “The” == “the” == “THE” → *lowercase / upper case*
- Punctuation , ! . ? → . “ ‘ « » ’ ” → ∅ ... !! ??? → .
- Numbers 1 2 3 5 8 → ∅ 3.14159 90210 → *as is/<NUMBER>*
- Special characters ∇ $ € § ¶ ** → ∅
- Special words 😊 #nlp → :happy: #nlp

```
corpus = 'Who ❤️ "word embeddings" in 2020? I do!!!'

data = re.sub(r'[,!?;-]+', '.', corpus)
data = nltk.word_tokenize(data)  # tokenize string to words
data = [ ch.lower() for ch in data
         if ch.isalpha()
         or ch == '.'
         or emoji.get_emoji_regexp().search(ch)
       ]
```

→ ['who', '❤️', 'word', 'embeddings', 'in', '.', 'i', 'do', '.']

SLIDING WINDOW OF WORDS IN PYTHON

```python
def get_windows(words, C):
    i = C
    while i < len(words) - C:
        center_word = words[i]
        context_words = words[(i - C):i] + words[(i+1):(i+C+1)]
        yield context_words, center_word
        i += 1
```

I	am	happy	because	I	am	learning
0	1	2	3	4	5	6

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Corpus I am happy because I am learning

Vocabulary am, because, happy, I, learning

One-hot vector

$$
\begin{array}{r|c|c|c|c|c|}
 & \text{am} & \text{because} & \text{happy} & \text{I} & \text{learning} \\
\text{am} & 1 & 0 & 0 & 0 & 0 \\
\text{because} & 0 & 1 & 0 & 0 & 0 \\
\text{happy} & 0 & 0 & 1 & 0 & 0 \\
\text{I} & 0 & 0 & 0 & 1 & 0 \\
\text{learning} & 0 & 0 & 0 & 0 & 1 \\
\end{array}
$$

VOCABULARY – ONE HOT ENCONDING

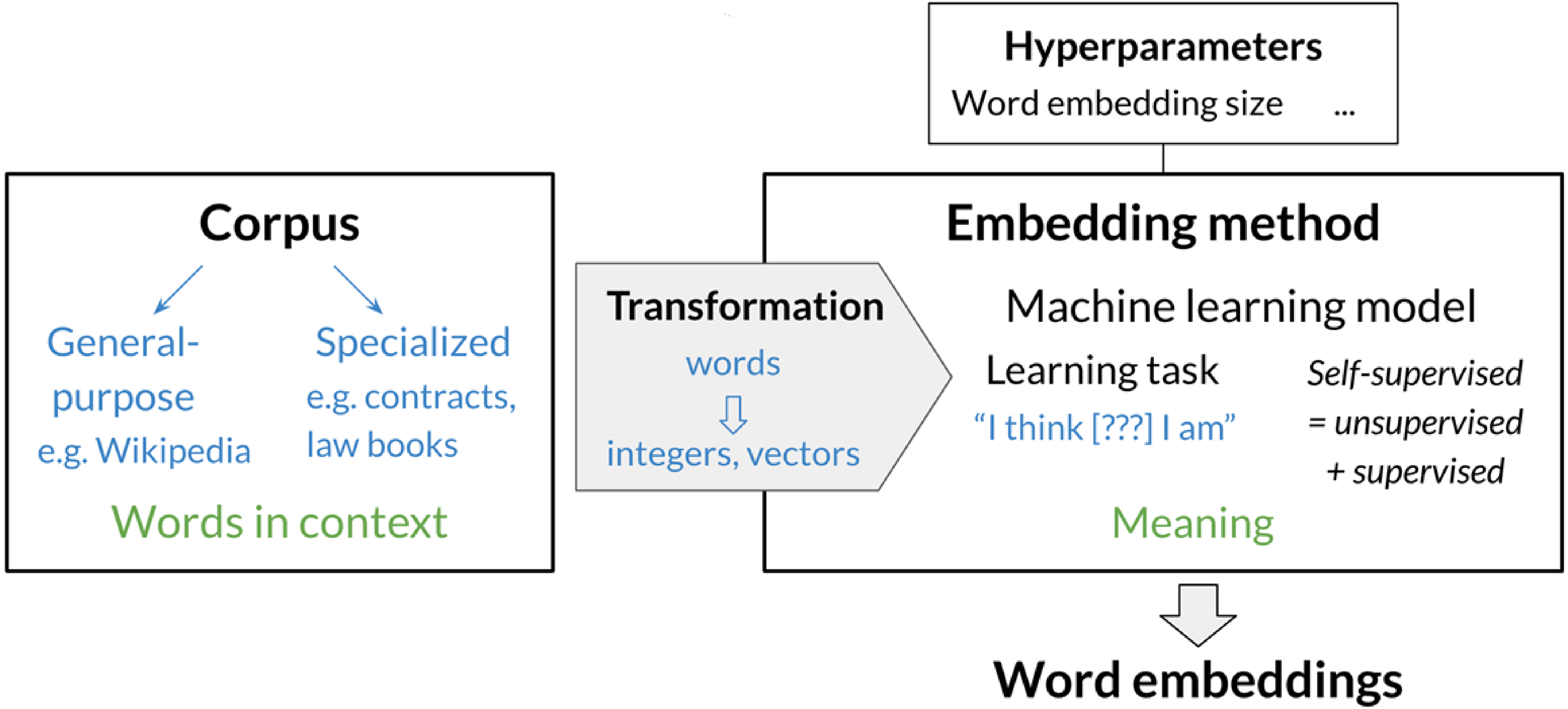
CONTEX WORDS LIKE SENTENCES IN CORPUS

$$
\left[
\begin{array}{r}
\\ \text{am} \\ \text{because} \\ \text{happy} \\ \text{I} \\ \text{learning}
\end{array}
\begin{array}{c} \text{I} \\ \begin{pmatrix}0\\0\\0\\1\\0\end{pmatrix} \end{array}
+
\begin{array}{c} \text{am} \\ \begin{pmatrix}1\\0\\0\\0\\0\end{pmatrix} \end{array}
+
\begin{array}{c} \text{because} \\ \begin{pmatrix}0\\1\\0\\0\\0\end{pmatrix} \end{array}
+
\begin{array}{c} \text{I} \\ \begin{pmatrix}0\\0\\0\\1\\0\end{pmatrix} \end{array}
\right] / 4 =
\begin{array}{c} \text{I am because I} \\ \begin{pmatrix}0.25\\0.25\\0\\0.5\\0\end{pmatrix} \end{array}
$$

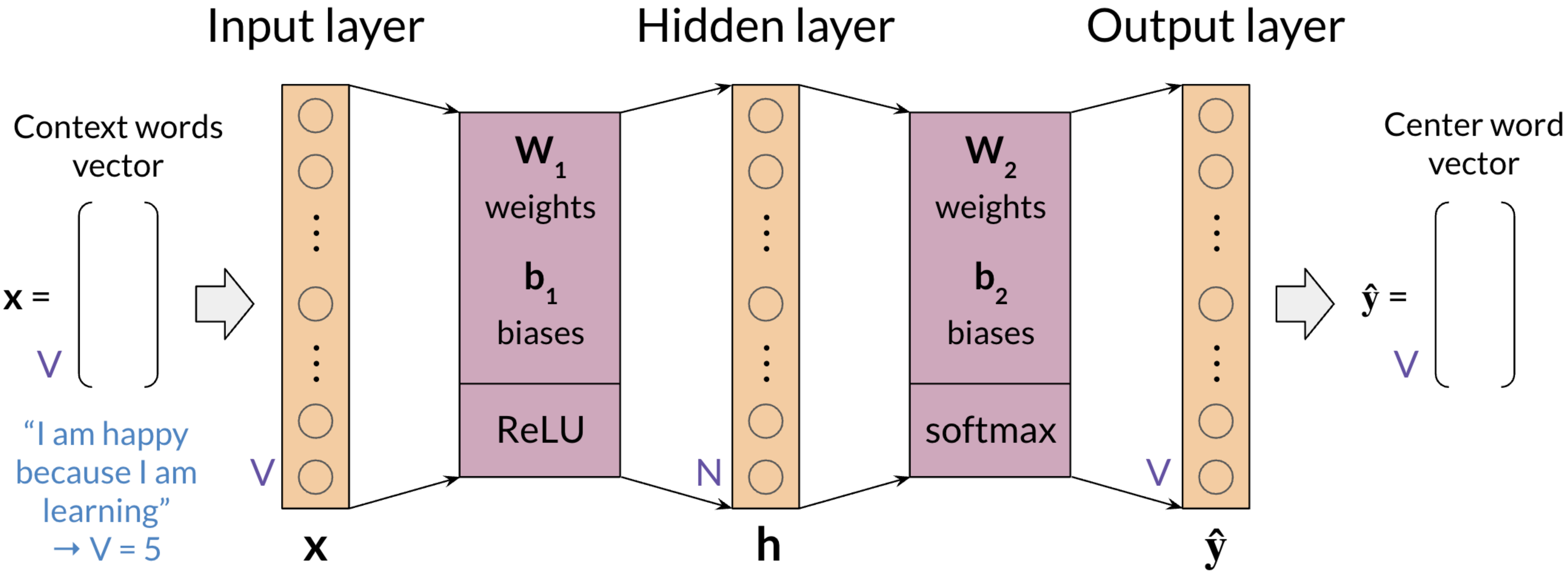
Minable view in the data scientist context

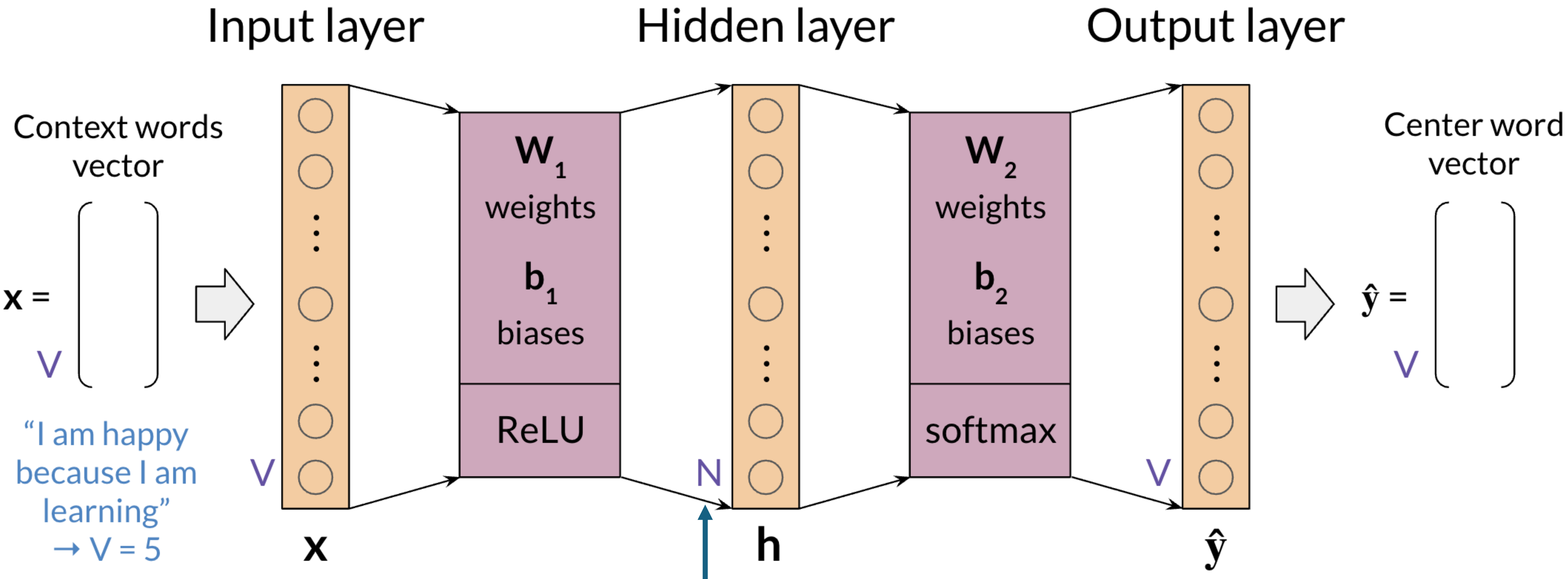
	x		y
Context words	Context words vector	*Center word*	Center word vector
I am because I	[0.25; 0.25; 0; 0.5; 0]	*happy*	[0; 0; 1; 0; 0]
am happy I am	[0.5; 0; 0.25; 0.25; 0]	*because*	[0; 1; 0; 0; 0]
happy because am learning	[0.25; 0.25; 0.25; 0; 0.25]	*I*	[0; 0; 0; 1; 0]

HOW TO CREATE WORD EMBEDDINGS?



ARCHITECTURE FOR THE CBOW MODEL





Hyperparameters:
N = Word embeddings size